

Hariprashad Ravikumar

PhD Candidate specializing in Physics-Based Modeling, Predictive Analytics, and AI for Science

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Location: San Jose, CA

Experience

- HAMR Modeling and Simulation Intern**, Western Digital, San Jose, CA *(May 2026 – Aug 2026)*
 - Built and shipped a full-stack physics simulation web app (Python, Dash, Plotly) that lets engineers run experiments computationally instead of on physical hardware; used daily in production by 40+ senior R&D engineers and subject-matter experts across the company's US and Japan sites
 - Added real-time parameter sweeps, side-by-side run comparison, and one-click auto-export to PowerPoint/Excel; built client-side state management with IndexedDB API for persistent, cross-tab data handoff
 - Packaged the tool as a modular, PyPI-installable Python library with a clean API; deployed on Kubernetes via a Jenkins CI/CD pipeline and integrated it into a broader internal engineering platform alongside related analysis tools
 - Derived a closed-form analytical model of magnetic grain switching behavior from first principles, combining thermal-activation and coherent-rotation theory to predict switching time, signal noise, and error probability across repeated write cycles; validated against Monte Carlo simulation and experimental lab data, with core findings in preparation for peer-reviewed publication
 - Applied statistical modeling and curve fitting to identify causal drivers of storage performance, letting engineers predict and reduce data-integrity errors computationally, removing a hardware-test step from the R&D loop
 - Presented results at PhD Expo 2026 (50+ attendees) and multiple internal engineering reviews (30+ attendees each), including a cross-site review with the company's Japan R&D team
- Graduate Research Assistant**, NM State University, Las Cruces, NM *(Aug 2021 – Present)*

PhD Project: Lattice Quantum Chromodynamics & Machine Learning Approaches

 - Generated 30,000+ high-fidelity synthetic data points by solving Partial Differential Equations with large-scale Hybrid Monte Carlo simulations (Markov Chain stochastic process) and built an end-to-end AI for Science pipeline to model the underlying physics, achieving over 98% predictive accuracy with symbolic regression machine learning.
 - Engineered parallelized Bayesian inference pipelines across multi-CPU architectures to fit non-linear, high-dimensional models; utilized automatic differentiation and full covariance matrices to ensure exact error propagation and numerical stability in multi-stage workflows (jackknife resampling)

Independent Collaborations

- Los Alamos National Laboratory** - Lattice Quantum Chromodynamics *(May 2024 – Present)*
 - Accelerated multi-terabyte scientific calculations by developing and optimizing parallelized C++ CUDA kernels for GPU-accelerated HPC clusters (NERSC Perlmutter) to reduce runtime for large-scale Hamiltonian Monte Carlo simulations.
 - Managed and executed 75,000+ CPU/GPU compute hours by designing and deploying custom SLURM workflows for large-scale job orchestration
- North Carolina State University** - Mathematical Physics *(Dec 2020 - Present)*
 - Analyzed algebraic symmetry constraints in mathematical physics using Mathematica-based symbolic computation on HPC clusters.

Technical Projects

- Cost-Aware Multi-Agent LLM Router** GitHub | Live Demo
 - Cut LLM inference cost 93.9% versus always using the top-tier model, with no statistically distinguishable accuracy loss on a 50-prompt disjoint holdout, by designing a calibrated confidence router (logistic

regression over logprob uncertainty, self-consistency dispersion, and embedding-distance features) that escalates only the prompts likely to need a costlier model

- Validated router confidence for honesty as well as raw accuracy, using Expected Calibration Error and Brier score, then deployed it as a production FastAPI service with Postgres decision logging, Redis caching, and an async circuit breaker on Google Cloud Run

2. AI-DataScience-Lab: Full-Stack Data Product

GitHub | Live App

- Prototyped and deployed a scalable full-stack data tool (Python, Flask, Azure) to deliver statistical analysis results to users, integrating the OpenAI API to auto-generate narrative summaries of those results.
- Implemented an end-to-end statistical analysis pipeline using Python (Pandas) for data cleaning, statistical modeling (Scikit-learn) for regression, and Matplotlib for data visualization.

Selected Publications

Google Scholar

1. C.-R. Ji and **H. Ravikumar**, "*Interpolating conformal algebra in $(1+1)$ dimensions*," **Physical Review D** 113, 096018 (May 2026). DOI: 10.1103/prlq-4j1l.

Technical Skills

Methods & HPC	Parallel Computing (GPU, MPI), Numerical Methods (PDEs, Monte Carlo, Regression)
Programming	Python, C++, CUDA, Bash, SQL
ML & APIs	PyTorch, TensorFlow, Scikit-learn, Pandas, Dash, Plotly, Flask, FastAPI, Numba
Cloud & MLOps	Azure, AWS (S3, Lambda), Docker, Kubernetes, Jenkins, CI/CD, Git, SLURM

Education

PhD in Physics, New Mexico State University, USA

Aug 2021 – Dec 2026 (expected)

MS in Physics, New Mexico State University, USA

Aug 2021 – May 2024

Certifications

- Fundamentals of Accelerated Computing with CUDA Python by NVIDIA

Awards

- **2025 NMC Collaboration Grant**, awarded by the New Mexico Consortium to conduct my independent research project in collaboration with scientists at Los Alamos National Laboratory
- **2023 George and Barbara Goedecke Physics Excellence Fund Scholarship**, awarded by the NMSU Physics Department

Leadership

- **Vice President**, Physics Graduate Student Organization (NMSU) *Aug 2025 – Present*
Organized professional development events and served as the primary liaison between 40+ graduate students and faculty.

Relevant Graduate Coursework

- Quantum Computing, Advanced Computational Physics, Statistical Mechanics, Electromagnetic Theory I & II